Trajectory-Selective Distillation for Retain-Set-Free Masked Diffusion Language Model Unlearning

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Abstract

We study retain-set-free machine unlearning for masked diffusion language models (MDLMs). The central design principle is to treat a token prediction under a particular mask configuration as the atomic denoising decision. We directly modify the frozen teacher distribution only at answer-bearing denoising positions, while self-distilling the original teacher at all remaining masked positions. To account for the bidirectional generation dynamics of MDLMs, the forgetting signal is applied across partially masked answer states rather than only at a fully masked response. This document gives the notation and algorithm for the resulting trajectory-selective distillation objective.

1 Problem Setup

Let p_{θ_0} be a pretrained or instruction-tuned MDLM and let $\mathcal{D}_{\mathrm{f}} = \{(x, y, \mathcal{A})\}$ be a forget set. Here, x denotes the conditioning prompt, $y = (y_1, \ldots, y_L)$ denotes the response, and $\mathcal{A} \subseteq \{1, \ldots, L\}$ contains the answer-bearing token positions whose information should be forgotten. We assume no retain examples are available during unlearning.

An MDLM operates on partially masked sequences. For a mask set $\mathcal{M} \subseteq \{1, \ldots, L\}$, let $y_{\mathcal{M}}$ denote the corrupted response obtained by replacing y_i with the mask token for every $i \in \mathcal{M}$. At a masked position $i \in \mathcal{M}$, the model predicts

$$p_\theta(\cdot \mid x, y_{\mathcal{M}}, \mathcal{M}). \tag{1}$$

The mask set is therefore part of the denoising state: the same answer token can be predicted under many different amounts of visible left and right context.

2 Notation

3 Trajectory-Selective Distillation

3.1 Answer-stratified mask sampling

Standard diffusion corruption may produce many states in which the answer is fully visible and hence provides no direct forgetting signal. We instead sample mask states from a distribution $Q_{\mathcal{A}}$ satisfying

$$\mathcal{M} \cap \mathcal{A} \neq \varnothing. \tag{2}$$

For a multi-token answer, the sampler should cover different numbers of visible answer tokens. A simple implementation first samples

$$k \sim \mathrm{Uniform}\{1, \ldots, |\mathcal{A}|\}, \tag{3}$$

then masks k uniformly chosen positions from $\mathcal{A}$. The remaining non-answer positions are masked according to the backbone's ordinary diffusion corruption distribution. This exposes the model to both fully masked and partially revealed answer states.

Table 1: Notation used by trajectory-selective distillation.

Symbol	Meaning
$\mathcal{D}_{\mathrm{f}}$	Forget dataset containing prompt–response pairs and answer-bearing spans.
(x, y)	Forget prompt x and response $y = (y_1, \ldots, y_L)$.
$\mathcal{A}$	Set of answer-bearing token positions in y.
$\mathcal{M}$	Set of response positions masked in the current denoising state.
$y_{\mathcal{M}}$	Partially masked response associated with mask set $\mathcal{M}$.
$Q_{\mathcal{A}}(\mathcal{M} \mid \mathcal{A})$	Answer-stratified mask distribution, constrained so that $\mathcal{M} \cap \mathcal{A} \neq \varnothing$.
θ_0	Frozen original MDLM parameters; used as the self-distillation teacher.
θ	Trainable MDLM parameters, initialized from θ_0.
$p^0_{\mathcal{M},i}$	Frozen-teacher distribution $p_{\theta_0}(\cdot \mid x, y_{\mathcal{M}}, \mathcal{M})$ at masked position i.
$\mathcal{B}_{\mathcal{M},i}$	Set of vocabulary tokens to demote at an answer-bearing decision.
$q_{\mathcal{M},i}$	Edited teacher target distribution at state $(\mathcal{M}, i)$.
$w_{\mathcal{M},i}$	Optional per-decision weight used to balance answer and non-answer masked positions.
$\mathcal{V}$	Vocabulary.

3.2 Selective teacher editing

For every masked position $i \in \mathcal{M}$, compute the frozen-teacher distribution

$$p^0_{\mathcal{M},i}(v) = p_{\theta_0}(v \mid x, y_{\mathcal{M}}, \mathcal{M}), \qquad v \in \mathcal{V}. \tag{4}$$

We then construct a single target distribution $q_{\mathcal{M},i}$ according to whether the current denoising decision predicts an answer-bearing position.

For a non-answer position, we preserve the original model exactly:

$$q_{\mathcal{M},i}(v) = p^0_{\mathcal{M},i}(v), \qquad i \notin \mathcal{A}. \tag{5}$$

For an answer-bearing position, let $\mathcal{B}_{\mathcal{M},i}$ be the set of vocabulary items to demote. The minimal version uses only the ground-truth forgotten token,

$$\mathcal{B}_{\mathcal{M},i} = \{y_i\}. \tag{6}$$

The edited teacher removes the demoted probability mass and renormalizes the remaining distribution:

$$q_{\mathcal{M},i}(v) = \begin{cases} 0, & v \in \mathcal{B}_{\mathcal{M},i}, \\ \dfrac{p^0_{\mathcal{M},i}(v)}{1 - \sum_{u \in \mathcal{B}_{\mathcal{M},i}} p^0_{\mathcal{M},i}(u)}, & v \notin \mathcal{B}_{\mathcal{M},i}. \end{cases} \tag{7}$$

A belief-demotion extension can enlarge $\mathcal{B}_{\mathcal{M},i}$ to include a small high-probability set under the frozen teacher, but this is not required by the core method.

3.3 Unified objective

The trainable model is fitted to the selectively edited teacher using one KL objective:

$$\boxed{\mathcal{L}_{\mathrm{TSD}}(\theta) = \mathbb{E}_{(x,y,\mathcal{A}) \sim \mathcal{D}_{\mathrm{f}}} \mathbb{E}_{\mathcal{M} \sim Q_{\mathcal{A}}(\cdot \mid \mathcal{A})} \left[\sum_{i \in \mathcal{M}} w_{\mathcal{M},i} D_{\mathrm{KL}}(q_{\mathcal{M},i} \,\|\, p_\theta(\cdot \mid x, y_{\mathcal{M}}, \mathcal{M})) \right].} \tag{8}$$

A simple balanced weighting scheme assigns half of the total weight to answer-bearing masked decisions and half to non-answer masked decisions:

$$\sum_{i \in \mathcal{M} \cap \mathcal{A}} w_{\mathcal{M},i} = \frac{1}{2}, \qquad \sum_{i \in \mathcal{M} \setminus \mathcal{A}} w_{\mathcal{M},i} = \frac{1}{2}, \tag{9}$$

Algorithm 1 Trajectory-Selective Distillation (TSD)

Require: Frozen MDLM p_{θ_0}; forget set $\mathcal{D}_{\mathrm{f}}$; trainable parameters $\theta \leftarrow \theta_0$; answer-stratified mask sampler $Q_{\mathcal{A}}$; learning rate η.

for each optimization step do
  Sample a minibatch $\mathcal{B} \subset \mathcal{D}_{\mathrm{f}}$.
  Initialize minibatch loss $\mathcal{L} \leftarrow 0$.
  for all $(x, y, \mathcal{A}) \in \mathcal{B}$ do
    Sample $\mathcal{M} \sim Q_{\mathcal{A}}(\cdot \mid \mathcal{A})$ such that $\mathcal{M} \cap \mathcal{A} \neq \varnothing$.
    Construct the partially masked response $y_{\mathcal{M}}$.
    Compute frozen-teacher distributions $p^0_{\mathcal{M},i} \leftarrow p_{\theta_0}(\cdot \mid x, y_{\mathcal{M}}, \mathcal{M})$ for all $i \in \mathcal{M}$.
    for all $i \in \mathcal{M}$ do
      if $i \in \mathcal{A}$ then
        Set $\mathcal{B}_{\mathcal{M},i} \leftarrow \{y_i\}$.
        Form $q_{\mathcal{M},i}$ by removing $\mathcal{B}_{\mathcal{M},i}$ from $p^0_{\mathcal{M},i}$ and renormalizing as in Eq. (7).
      else
        Set $q_{\mathcal{M},i} \leftarrow p^0_{\mathcal{M},i}$.
      end if
    end for
    Compute student distributions $p_\theta(\cdot \mid x, y_{\mathcal{M}}, \mathcal{M})$.
    $\mathcal{L} \leftarrow \mathcal{L} + \sum_{i \in \mathcal{M}} w_{\mathcal{M},i} D_{\mathrm{KL}}(q_{\mathcal{M},i} \| p_\theta(\cdot \mid x, y_{\mathcal{M}}, \mathcal{M}))$.
  end for
  Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$.
end for
return unlearned parameters θ.

whenever both groups are nonempty. If one group is empty, its mass is assigned to the other group.

For $i \in \mathcal{A}$ and the minimal demotion set in Eq. (6), $q_{\mathcal{M},i}(y_i) = 0$. Thus the derivative of the KL loss with respect to the student's target-token logit $z_{\theta,\mathcal{M},i}(y_i)$ is

$$\frac{\partial}{\partial z_{\theta,\mathcal{M},i}(y_i)} D_{\mathrm{KL}}(q_{\mathcal{M},i} \| p_\theta) = p_\theta(y_i \mid x, y_{\mathcal{M}}, \mathcal{M}) > 0, \tag{10}$$

whenever the model assigns nonzero probability to the forgotten token. Gradient descent therefore directly reduces its logit at every sampled answer-bearing denoising state.

4 Minimal Experimental Ablation

The core mechanism can be isolated using four variants:

1. Single-state demotion: demote answer tokens only when the entire answer span is masked.
2. Pathwise demotion: apply the same target demotion over fully and partially masked answer states.
3. Pathwise demotion + TSD: additionally self-distill the frozen teacher at all non-answer masked positions using Eq. (5).
4. Belief demotion: replace $\{y_i\}$ by a small teacher-defined high-probability demotion set to test whether broader belief suppression reduces paraphrase leakage.

The first diagnostic is whether pathwise demotion suppresses exact answer reconstruction under all levels of partial answer visibility. Only after this condition is met should retain utility and semantic paraphrase leakage be used to evaluate the preservation and belief-demotion components.